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Toxicity of new generation flame retardants to *Daphnia magna*

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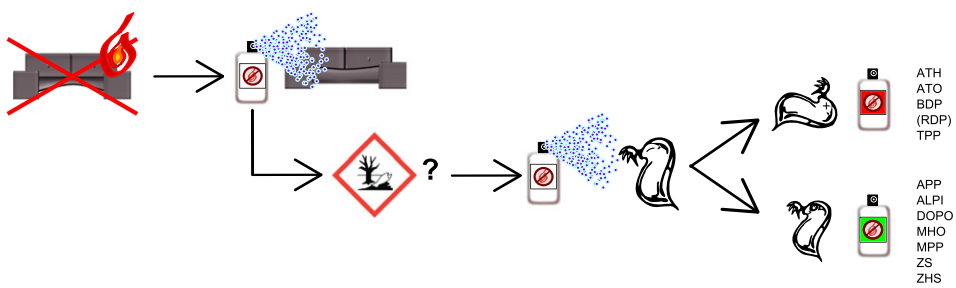
HIGHLIGHTS

- Ecotoxicity of many halogen-free flame retardants is unknown.
- Missing toxicity data of HFFRs were provided and inconsistencies were clarified.
- Seven tested HFFRs were not acutely toxic to *Daphnia magna*.
- TPP and the reference compound TBBPA were highly toxic to *Daphnia magna*.
- Best candidates for BFR replacements are: APP, ALPI, DOPO, MHO, MPP, ZHS & ZS.

GRAPHICAL ABSTRACT



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ABSTRACT

There is a tendency to substitute frequently used, but relatively hazardous brominated flame retardants (BFRs) with halogen-free flame retardants (HFFRs). Consequently, information on the persistence, bioaccumulation and toxicity (PBT) of these HFFRs is urgently needed, but large data gaps and inconsistencies exist. Therefore, in the present study the toxicity of a wide range of HFFRs to the water flea *Daphnia magna* was investigated. Our results revealed that four HFFRs were showing no effect at their S_w (saturated water concentration) and three had a low toxicity ($EC_{50} > 10$ mg L^{-1}), suggesting that these compounds are not hazardous. Antimony trioxide had a moderate toxicity ($EC_{50} = 3.01$ mg L^{-1}, 95% CL: 2.76–3.25) and triphenyl phosphate and the brominated reference compound tetra bromobisphenol A were highly toxic to *D. magna* ($EC_{50} = 0.55$ mg L^{-1}, 95% CL: 0.53–0.55 and $EC_{50} = 0.60$ mg L^{-1}, 95% CL: 0.24–0.97 respectively). Aluminum trihydroxide and bisphenol A bis(diphenyl phosphate) caused limited mortality at S_w (26 and 25% respectively) and have a low solubility (<10 mg L^{-1}). Hence, increased toxicity of these compounds may be observed when for instance decreasing pH could increase solubility. By testing all compounds under identical conditions we provided missing insights in the environmental hazards of new generation flame retardants and propose as best candidates for BFR replacements: APP, ALPI, DOPO, MHO, MPP, ZHS and ZS.

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Abbreviations: ALPI, aluminum diethyl phosphinate; APP, ammonium polyphosphate; ATH, aluminum trihydroxide; ATO, antimony trioxide; BDE, brominated diphenylether; BDP, bisphenol A bis(diphenyl phosphate); BFR, brominated flame retardant; CAS no, chemical abstracts service number; DOPO, 9,10-dihydro-9-oxa-10-phosphaphenanthrene; dTPP, deuterated triphenyl phosphate; EC_{50}, concentration that causes an effect (here immobility) to 50% of the test population; ECHA, European chemicals agency; EP, European parliament; FR, flame retardant; HBCD, hexabromocyclododecane; HFFR, halogen-free flame retardant; HOC, hydrophobic organic contaminant; ICP-AES, inductively coupled plasma (coupled to) atomic emission spectroscopy; ISO, international organization for standardization; LC-MS/MS, high performance liquid chromatography (coupled to) tandem mass spectrometry; MHO, magnesium hydroxide; MOPS, 3-(N-morpholino)propanesulfonic acid; MPP, melamine polyphosphate; OECD, organization for economic co-operation and development; PBB, polybrominated biphenyls; PBDE, polybrominated biphenyl ether; PBT, persistence, bioaccumulation and toxicity; RDP, resorcinol bis(diphenyl phosphate); REACH, registration, evaluation, authorization and restriction of chemical substances; S_w, saturated water concentration; TBBPA, tetra bromobisphenol A; TOC, total organic carbon; TPP, triphenyl phosphate; ZHS, zinc hydroxystannate; ZS, zinc stannate.

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1. Introduction

Chemical additives known as flame retardants (FRs) are incorporated into a wide range of polymers to fulfill regulatory requirements on flame retardancy. Brominated flame retardants (BFRs) are frequently used because they have a low impact on the polymer's characteristics, are very effective in relatively low amounts compared to other FRs (Alaee et al., 2003), and are relatively cheap (Birnbaum and Staskal, 2004). In 2004, BFRs accounted for about 21% of the total world production of FRs (SRI Consulting (SRIC), 2004). Many BFRs, however, have unintended negative effects on the environment and human health. Some are very persistent (Robrock et al., 2008), some bioaccumulate in aquatic and terrestrial food chains (Boon et al., 2002), and some show serious adverse effects such as endocrine disruption (Meerts et al., 2001).

Concerns about the persistence, bioaccumulation, and toxicity (PBT) of BFRs have led to a ban on the production and use of many of these compounds, such as polybrominated biphenyls (PBBs) and several brominated diphenylethers (BDEs) (European Parliament (E.P.), 2002; OSPAR, 2001 (2004 updated)). Hence, there is growing need to substitute BFRs with alternative halogen-free flame retardants (HFFRs), and several furniture manufacturers have already voluntarily replaced BFRs with alternative HFFRs (Betts, 2007). HFFRs can be divided into several categories, the most important ones being: inorganic flame retardants and synergists (mostly used for electronics and electrical equipment), organophosphorus compounds and their salts (housings of consumer products), nitrogen-based organic flame retardants (electronics and electrical equipment) and intumescent systems (textile coatings). Because of the need for BFR substitution, many of these HFFRs are already being marketed, although their environmental behavior and toxicological properties are known to only a limited extent and their potential impact on the environment cannot yet be properly assessed. As a result, there is urgent need for information on the PBT properties of HFFRs. Reviewing the publicly available ecotoxicity data of HFFRs we identified large data gaps and inconsistent observations on the properties of individual compounds (Waaijers et al., 2013). Therefore, the aim of this study was to generate reliable toxicity data for a selection of HFFRs that are potential replacements for BFRs in polymers.

For this study, twelve HFFRs were selected: antimony trioxide (ATO), aluminum trihydroxide (ATH), magnesium hydroxide (MHO), zinc hydroxystannate (ZHS) and zinc stannate (ZS) (inorganic flame retardants and synergists); aluminum diethylphosphinate (ALPI), bisphenol A bis(diphenylphosphate) (BDP), 9,10-dihydro-9-oxa-10-phosphaphenanthrene (DOPO), resorcinol bis(diphenylphosphate) (RDP) and triphenylphosphate (TPP) (organophosphorus compounds and salts), melamine polyphosphate (MPP) (nitrogen based organic flame retardant); ammonium polyphosphate (APP) (intumescent systems). Tetrabromobisphenol A (TBBPA) was tested as well, as a BFR reference compound.

In order to assess the toxicity of this large group of compounds, we performed the standardized OECD 202 acute daphnid immobility tests (Organisation for Economic Co-operation and Development (OECD), 2004). The data obtained from these experiments (EC_{50} values) were classified based on the REACH system (European Union, 2006, 2008), meaning that we assigned the categories "high", "moderate" and "low" toxicity to the data ($EC_{50} < 1$ mg L^{-1}, 1–10 mg L^{-1} & >10 mg L^{-1} respectively). In this way we were able to rapidly screen a large selection of new generation flame retardants and compare their toxicity to the one of the most commonly used test organism *D. magna*.

2. Materials and methods

2.1. Test organism and culture conditions

We chose the fresh water filter feeder *Daphnia magna* Straus to screen an array of flame retardants for aquatic toxicity. This water flea is frequently used as test organism in ecotoxicity studies due to several benefits, including its key role in the pelagic food webs of temperate regions, parthenogenetic reproduction (which excludes genetic variation) and ease of handling (Adema, 1978; Burns, 1969). The *D. magna* neonates (younger than 24 h, clone 4) used in this study were obtained from Grontmij Aquasense (Amsterdam, the Netherlands). The daphnid cultures were kept for three weeks, after which new cultures were started with neonates (younger than 24 h). The cultures were maintained in 4–4.5 L Elendt M4 medium in glass aquaria, corresponding to a minimum of 30 mL per adult (OECD, 2004). The medium had a pH of 7.8 ± 0.5, a conductivity of 50–80 μS/mm and was kept under continuous aeration. Cultures were maintained under a light-dark regime of 16:8 h (twilight zone of 30 min) and at a temperature of 20 ± 1 °C. The medium was renewed two times a week and simultaneously neonates were removed. The daphnids were fed five days a week with a suspension of the alga *Scenedesmus subspicatus* originating from a batch culture in CP-medium (NPR 6505, 1994). The algal culture was kept in a climate room at 20 ± 1 °C under continuous light and aeration. Every two weeks, algae were harvested by filtration (0.45 μm). The supernatant was removed and the algae were resuspended in Elendt M 4 medium (stored at 4 °C in the dark until feeding). The cell density was verified with a spectrophotometer (Hachlange Dr2800) and total organic carbon (TOC) with a TOC analyser (TOC-V cph, Shimadzu). The density of the food suspension corresponded to 3×10^9 cells L^{-1} and about 65 mg carbon L^{-1}. The culture was fed 69 mL (days 1–2), 102 mL (days 3–7) and 139 mL (day 8 and further) of algae suspension per day.

At regular intervals (about every three months), acute toxicity tests were performed with the reference toxicant $K_2Cr_2O_7$ to check whether the sensitivity of the *D. magna* culture was within the limits (EC_{50}, 24 h = 0.6–2.1 mg L^{-1}) as set by the guideline (OECD, 2004).

2.2. Test compounds: halogen-free flame retardants

The HFFRs studied were selected based on the most important current applications as BFR replacements in polymers. These included six organophosphates: aluminum diethyl phosphinate (CAS no. 225789-38-8, 98.5%, Clariant), bisphenol A bis(diphenyl phosphate) (polymer consisting of mostly n = 1–2, CAS no. 5945-33-5, ≥80%, ICL), 9,10-dihydro-9-oxa-10-phosphaphenanthrene (CAS no. 35948-25-5, ≥98%, KCCS), melamine polyphosphate (polymer consisting <n> = 50, CAS no. 218768-84-4, >99.5%, BTC U.K.), resorcinol bis(diphenyl phosphate) (polymer consisting of mostly n = 1–3, CAS no. 57583-54-7, 82%, ICL), triphenyl phosphate (CAS no. 115-86-6, 99%, Sigma-Aldrich) and six inorganic compounds: aluminum trihydroxide (CAS no. 21645-51-2, ≥80%, Merck), ammonium polyphosphate (polymer consisting of <n> = 1000, CAS no. 68333-79-9, 99.5%, Clariant), antimony trioxide (CAS no. 1309-64-4, 99.3%, Chemtura Belgium N.V.), magnesium hydroxide (MHO, CAS nr 1309-42-8, ≥99%, Fluka), zinc hydroxy stannate (ZHS, CAS no. 12027-96-2, 98.5%, William Blythe) and zinc stannate (ZS, CAS no. 12036-37-2, 99%, William Blythe). dTPP (98%, d-15, internal standard, product nr: 615218) was obtained from Sigma-Aldrich (Zwijndrecht, Netherlands). The chemical structures of the organophosphates are shown in Fig. 1. For BDP and RDP only low purity technical products (≥80%) were available. The toxicity of tetra bromobisphenol A (TBBPA, CAS no. 79-94-7, 97%, Sigma-Aldrich) was tested as a reference compound for brominated flame retardants.

2.3. Test solutions

Aqueous solubility data of the compounds were often lacking or inconsistent, covering a wide range of values (Waaijers et al., 2013). Therefore, we decided to first test the effect of saturated water solutions (ISO medium without additional buffer, pH = 7.5 ± 0.5, T = 20 °C ± 1, here defined as S_w) on the daphnids. To this purpose we stirred an excess of compound for 7 days in ISO

medium, (OECD, 2004) and then filtered the solution through glass fiber filters (GWSC, Millipore B.V., 0.45 μm). Next, this solution and a control (ISO medium) were tested for toxicity. The exceptions to this approach were TBBPA, TPP and ATH, which were spiked with methanol (0.08%) into ISO medium at the highest value reported in literature for S_w (respectively 4, 2 and 1.6 mg L^{-1} nominal). For these compounds a solvent control was included in the tests as well. For the compounds that caused more than 50% immobility of the daphnids at S_w, a dilution range was prepared (4 concentrations and a control) and tested in order to obtain concentration–response relationships and to derive EC_{50} values. The test concentrations for each compound are listed in the Supplementary data (Table S1).

2.4. Toxicity tests

To determine the acute toxicity of the HFFRs, *D. magna* were exposed to the selected compounds in 48 h immobility tests, in accordance with the OECD guideline 202 (2004), except where noted. Per concentration (S_w, dilution or control) four replicates were prepared. Each replicate consisted of a polypropylene tube filled with 40 mL of test solution (S_w, dilution or control). The tubes were randomly distributed in a climate controlled fume hood (20 ± 1 °C), with a light-dark regime of 16:8 h. The experiment was started by introducing 5 neonates (younger than 24 h) into each tube using a disposable transfer pipette. After 24 and 48 h, the number of animals not responding to gentle stimulation by tapping on the tube was scored. Physical–chemical parameters (hardness, oxygen level, temperature & pH) recommended by the guideline were measured (Table S2). For DOPO, the pH of the medium dropped significantly in the highest test concentrations (pH < 6) and therefore an additional test was performed with buffered medium. Test medium was buffered to pH 7.5 ± 0.3 with NaOH (120 mg L^{-1}) and 3-(N-morpholino)propanesulfonic acid (MOPS) (628 mg L^{-1}) according to De Schamphelaere et al. (2004).

2.5. Analysis of inorganic HFFRs and of ALPI, DOPO, MPP and TBBPA

The concentrations of HFFRs in the S_w solutions were measured after a week of stirring and subsequent filtration. This was done to determine the actual concentration at the start of the experiment and when necessary, to prepare dilution ranges. Additionally, for all treatments (S_w, dilution and control) samples were measured at the end of the toxicity tests (48 h). For samples containing inorganic HFFRs or ALPI, DOPO or MPP, compound concentrations of each treatment (S_w, dilution or control) were determined by sacrificing additional replicates containing no daphnids (one to three for each compound, details Table S1). Of each of these replicates 15 mL water was taken, acidified (0.4% nitric acid and 2.38 g cesium chloride L^{-1}, both obtained from Merck) and stored until analysis at 4 °C. Samples were measured using inductively coupled plasma coupled to atomic emission spectroscopy (ICP-AES, Optima 3000XL). The following elements were measured to calculate the concentrations: aluminum for ATH; antimony for ATO; magnesium for MHO; phosphorus for ALPI, APP, DOPO and MPP; and zinc for ZHS and ZS. The TBBPA samples were not analyzed and concentrations of TBBPA mentioned in this study are therefore nominal.

2.6. Analysis of BDP, RDP and TPP

Samples containing BDP, RDP or TPP (the remaining organic HFFRs) were stored until analysis at −18 °C. To determine the concentrations of BDP and RDP (S_w, dilution or control) three water subsamples (500 μL or 50 μL respectively) were taken from three of the four replicates per treatment, resulting in nine samples per treatment. This was done in order to check the homogeneity of the test solutions as RDP and BDP were observed to form emulsions. For TPP one water subsample of 500 μL was taken from three of the four replicates per treatment, resulting in three samples per treatment. Deuterated triphenyl phosphate (dTPP, d-15, product nr: 615218, 98%, Sigma-Aldrich) was used

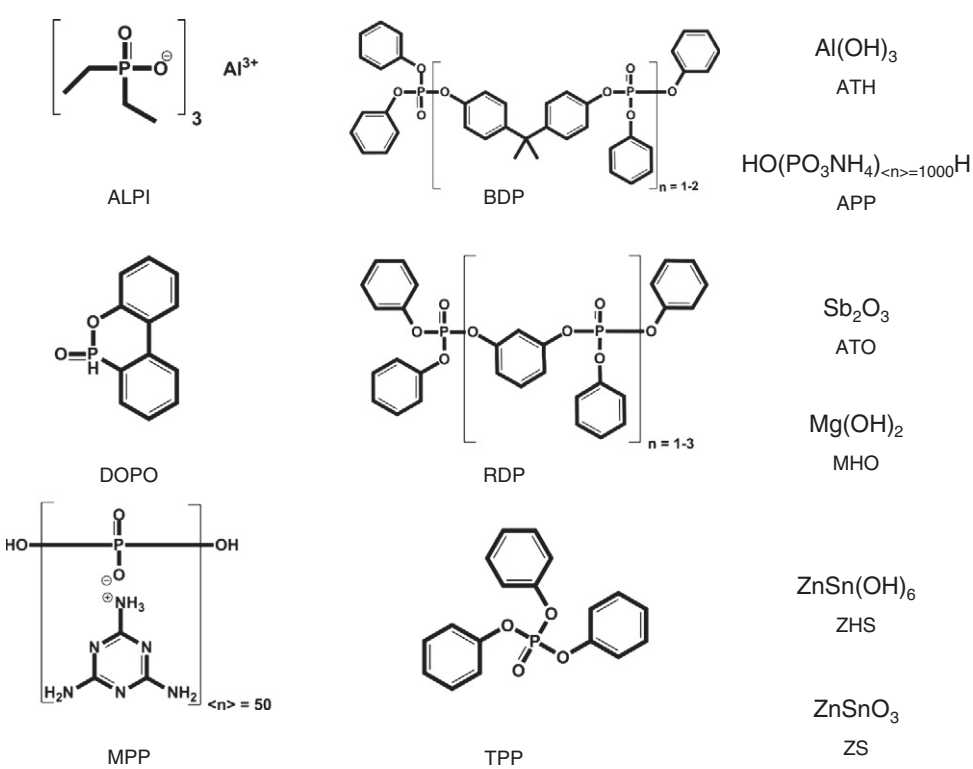

Fig. 1. Overview of the selected HFFRs for this study. Six (semi) organic compounds (schematic structure) and six inorganics (molecular formula) are shown.

as internal standard for the analysis of BDP, RDP and TPP. BDP and RDP consist primarily of their monomeric form (±80%) and measured concentrations were therefore monomer based. Solvents were supplied by Biosolve (Valkenswaard, The Netherlands). Ultra-pure water was used from Biosolve (ULC grade, Valkenswaard, The Netherlands) as well as from an ELGA water system (ELGA ultra-pure water system, Ubstadt-Weiher, Germany). Stock solutions were made for TPP, RDP, BDP (2000 mg L^{-1}), and dTPP (500 mg L^{-1}), the former three were used to prepare a mixed stock solution (200 mg L^{-1}). All stock solutions and calibration standards were prepared in acetonitrile.

The concentrations of BDP, RDP and TPP in the water samples were measured using high performance liquid chromatography coupled to tandem mass spectrometry (LC-MS/MS). Samples were quantified using dTPP as an internal standard (IS). Eluent A consisted of MeOH:H_2O 50:50 and eluent B of pure MeOH, both containing 5 mM ammonium acetate (pH 5). The gradient was as follows: 0 min (75% B), 11 min (100% B), 17 min (100% B), and 19 min (75% B). A C18 stationary phase column (Luna C18(2), 3 μm, 100 Å, 150 × 3.0 mm ID, Phenomenex, Torrance, CA, USA) was used. The LC system (Shimadzu, Kyoto, Japan) consisted of an autosampler (SIL-20A), binary pump (LC-20AD XR), column oven (CTO-20 AC) and system controller (CBM-20A). The LC was interfaced to a Sciex 4000 QTRAP MS/MS (Applied Biosystems, Toronto, Canada). For quantification two (precursor-product) transitions were measured in positive mode with multiple reaction monitoring (MRM). For BDP the transitions 693 → 367, 327 were measured, for RDP 575 → 481, 419, TPP 327 → 215, 153 and dTPP 342 → 243, 162. Analyst software (version 1.5.1) was used to operate the LC-MS/MS.

2.7. Data analysis

Concentration–response relationships and the corresponding 48-h EC_{50} values were calculated according to Haanstra et al. (1985) by fitting a logistic curve (Eq. (1)) through the percentage of mobility (100% − immobilization) against the HFFR concentration in the water.

$$y(x) = \frac{C}{1 + e^{b(\log_{10}x - \log_{10}a)}}. \quad (1)$$

In this equation $y(x)$ is the mobility at concentration x (%), a is the EC_{50} (mg L^{-1}), b is the slope of the curve, c is y(0), which equals the average mobility of the control and x is the concentration of FR in the water (mg L^{-1}). Data analyses were performed with SPSS software (V20.0.0) (IBM, 2011).

3. Results

Acute toxicity tests performed with the reference toxicant $K_2Cr_2O_7$ showed that the sensitivity of the *D. magna* culture was within the limits as set by the guideline (EC_{50}, 24 h = 0.6–2.1 mg L^{-1}, OECD, 2004). The brominated FR TBBPA was highly toxic to *D. magna* (EC_{50} = 0.60 mg L^{-1}, see Table 1) after 48 h, well within the ranges reported in literature (European Chemicals Agency (ECHA) Database original study, 1978, 2003; Liu et al., 2007). The physical–chemical parameters (hardness, oxygen level, temperature & pH, shown in Table S2) were within the recommended ranges and the mobility in the (solvent) controls was 90% or higher, conform the criteria of the guideline (OECD, 2004).

Several compounds were not acutely toxic, five of them (MPP, MHO, RDP, ZHS and ZS) showing no effect at S_w ($EC_{50} > S_w$) and two of them (ATH and BDP) showing limited immobility at S_w (25–26%, see Table 1). For the compounds that were toxic below their S_w, clear concentration–response relationships were observed as shown in Fig. 2. Corresponding parameters and confidence limits are reported in Table S3. From these concentration–response relationships EC_{50} values were derived according to Eq. (1) and these are listed in Table 1.

For ALPI, APP and DOPO a low acute toxicity was observed (EC_{50} > 10 mg L^{-1}, based on the REACH classification European Union, 2006, 2008). When the test medium of DOPO was buffered (pH = 7.5 ± 0.3) no effect was observed up to 289 mg L^{-1} (the highest concentration tested) confirming the low toxicity of DOPO. The toxicity of ATO was moderate (EC_{50} = 1–10 mg L^{-1}). TPP exerted a high toxicity after 48 h to *D. magna* (EC_{50} < 1 mg L^{-1}).

4. Discussion

In the present study the HFFRs tested were considered as good potential candidates for BFR replacement when they were not acutely toxic at their S_w ($EC_{50} > S_w$), since no effect observed at S_w suggests that the compound is non-hazardous. However, the solubilities of some of the HFFRs are so low that a slight change in solubility (for example due to decreasing pH Wood, 1985; Wren and Stephenson, 1991) could potentially affect their classification. If a compound has a low S_w value and causes limited immobility (and EC_{50} may even be > S_w), they could still be classified as highly or moderately toxic, if prevailing conditions in the environment increase solubility of the compound and causes the EC_{50} to be below S_w. In contrast, if S_w would decrease, no effect might be observed for such a compound. While a factor of ten or twenty is not considered as a high variability for S_w, this variability would clearly affect toxicity endpoints. Based on this argumentation, we considered only those HFFRs that did not cause any immobility at S_w or compounds with EC_{50} > 10 mg L^{-1} as truly non-hazardous and potential candidates for BFR replacement. In coming paragraphs each tested FR is evaluated and also the possible influence of pH on change in solubility and toxicity is taken into account whenever relevant.

The results of the present study revealed that MHO, MPP, ZHS and ZS exhibited no acute toxic effect at their S_w ($EC_{50} > S_w$). No significant influence of pH changes in the environment on the toxicity is expected for MHO and MPP as their solubility is very high and at these high concentrations no effects were observed. Up to pH 8 the Zn ion is relatively soluble (>10 mg L) (U.S. Environmental Protection Agency (EPA), 1973), but, as the concentrations of zinc in ZHS and ZS at S_w were much lower, this possibly indicates that these metal oxides do not ionize in water, which might be confirmed by their stability and low solubility (Waaijers et al., 2013).

The toxicity of RDP could not be properly evaluated as it formed an emulsion in the test solution, seriously hampering a reliable toxicity

Table 1
Acute toxicity of flame retardants to *D. magna* expressed as 48 h EC_{50} values (mg L^{-1}).[a]

Compound	EC_{50} (48 h, mg L^{-1})	Toxicity classification (REACH based)
MPP	>S_w (2018)	n.a.
MHO	>S_w (439)	n.a.
RDP	>S_w (66)[b]	n.a.
ZHS	>S_w (2.69)	n.a.
ZS	>S_w (2.06)	n.a.
BDP	>S_w (0.21, 25% immobility)	n.a.
ATH	>S_w (0.065, 26% immobility)	n.a.
APP	730 (95% CL: 718–742)	Low
$DOPO_{unbuffered}$	240 (95% CL: 240–240)	Low
$DOPO_{buffered}$	>289	n.a.
ALPI	18.43 (95% CL: 15.10–21.77)	Low
ATO	3.01 (95% CL: 2.76–3.25)	Moderate
TBBPA	0.60 (95% CL: 0.24–0.97)[c]	High
TPP	0.55 (95% CL: 0.53–0.57)	High

[a] >S_w stands for higher than S_w, with the measured water concentration (equals the highest achieved test concentration) in brackets. For explanation of toxicity classification see text. n.a. stands for not applicable.
[b] RDP forms an emulsion, this is the average measured concentration (see Table S1).
[c] TBBPA samples were not analyzed, this represents a nominal concentration.

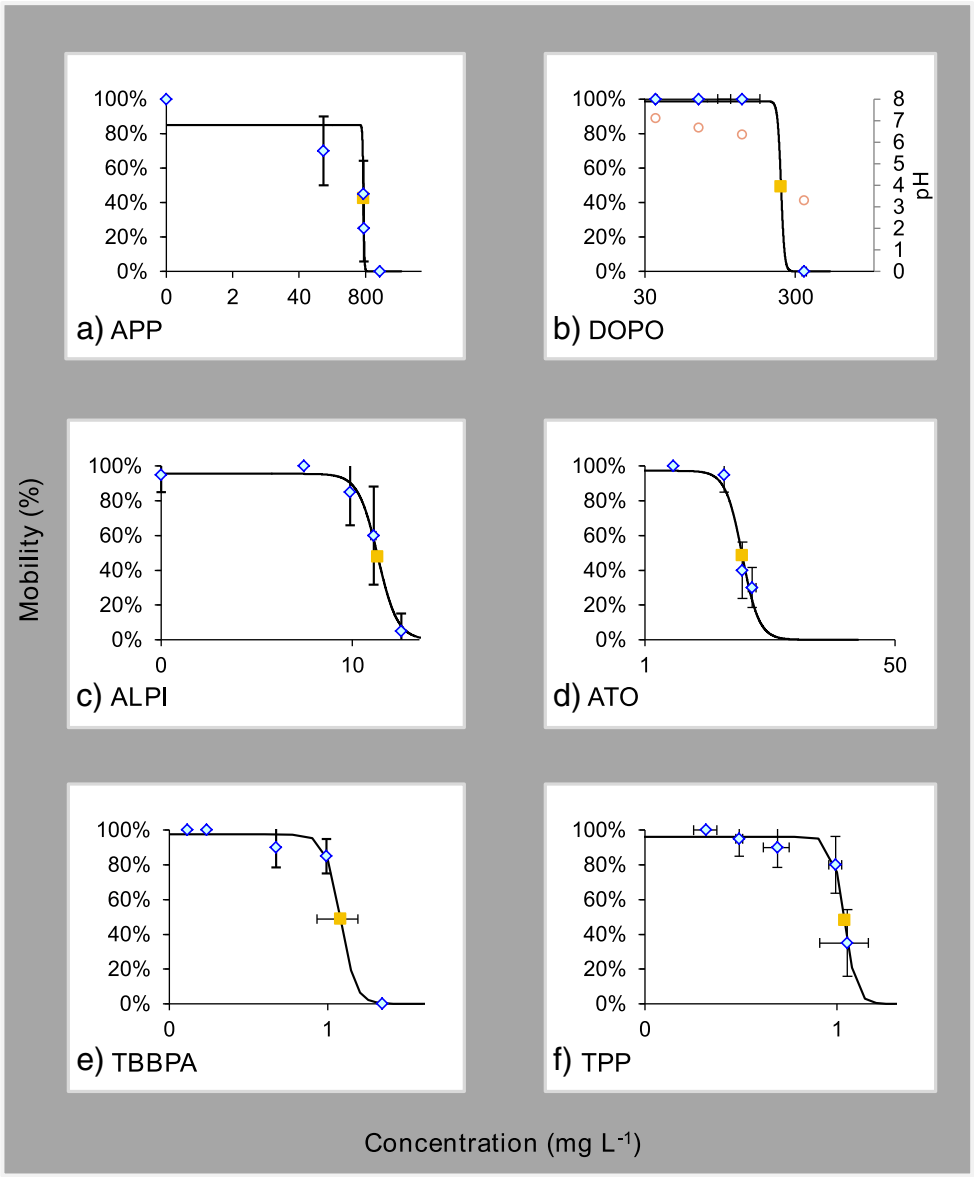

Fig. 2. Average (±stdev in y and for b–d & f ±stdev in x) mobility (◇, % of initial animals) of *Daphnia magna* (n = 4) after 48 h of exposure to HFFRs in ISO medium (mg L^{-1}). The EC_{50} is plotted as ■ (±stdev) and the logistics curve represents the fitted concentration–response relationship. For panel b) DOPO, the pH is plotted as ○. The brominated FR TBBPA served as a reference to literature (positive control).

assessment. The solutions obtained in this study after one week of stirring (emulsion; 66 ± 61 mg L^{-1}) exerted no adverse effect on the daphnids. However, pending an appropriate solution for this experimental artifact, RDP cannot as yet be identified as a first choice alternative FR. Moreover this technical RDP product has a low purity (~80%) and impurities are mostly the dimer form (~14% w/w), trimer (~2–3% w/w) and TPP (~3% w/w) (Riddell and Wellington Laboratories, 2012a, 7 March). For some hydrophobic organic contaminants (HOCs), such as polybrominated biphenyl ethers (PBDEs), a certain molecular weight cut off is observed for absorption across organisms' tissue (Leeuwen and Vermeire, 2007). Because of the high molecular weight of the dimer (823 g mol^{-1}) and trimer (1071 g mol^{-1}) we therefore expect that their uptake will be limited by their larger size compared to the monomer. However, as the toxicity mechanism of these compounds is still unknown, a purified standard should become available for testing to differentiate between the effects of the monomer and oligomers. Additionally, TPP might contribute to RDP toxicity (Table 1), especially at higher concentrations.

According to the present study, ATH and BDP would be less preferred alternative FRs because 25–26% immobility was already observed at their S_w, which was lower than 1 mg L^{-1}. For ATH two other studies reported an EC_{50} of 0.8 mg L^{-1} (*D. magna*) (U.S. Environmental Protection Agency (EPA), 2008) and of 2.6–3.5 mg L^{-1} (*Daphnia sp.*) (Illinois EPA, 2007). In the present study, we could not reproduce the same water saturation concentration for ATH and did not achieve concentrations higher than 0.1 mg L^{-1}. At this concentration 26% immobility was observed. ATH had a purity of (minimally) 80%, while the remaining impurity seems to be predominantly water (Merck Chemicals-Product Information (MSDS), 2007). Therefore, probably the toxicity observed was only caused by ATH and not by any impurities. As with the other tested inorganic FR, the solubility, complexation and speciation of ATH depend on the pH of the environment (Leeuwen and Vermeire, 2007). At the environmentally relevant pH range of about 6–8.5 (Hem and Geological Survey (U.S.), 1985; Michaud, 1991), the solubility of the total amount of aluminum is relatively low and estimated to be around 0.05 mg L^{-1} (Stumm and Morgan, 1996),

with its minimum around pH 6.5–7. The pH in this study (7.56 ±0.16) is well within the environmentally relevant range, showing that even around the lowest solubility of aluminum adverse effects on daphnids were observed.

As previously mentioned, for BDP we also observed toxic effects (25% immobility). The tested BDP technical product had a low purity (~81%) and impurities are mostly the dimer form (~16.5% w/w, 1059 g mol^{-1}) and TPP (~3% w/w) (Riddell and Wellington Laboratories, 2012b, 5 March). BDP toxicity could be caused by either the BDP monomer, dimer or TPP (Table 1). Therefore, as with RDP, a purified standard of BDP should become available for testing to distinguish between the effects of the different compounds present in the technical product.

For compounds that were toxic at levels below S_w, clear concentration–response relationships were observed, from which EC_{50} values were derived. APP and ALPI were classified as having a low toxicity because their EC_{50} values were above 10 mg L^{-1}. For DOPO also an EC_{50} was obtained, but in water the ring P – O bond of DOPO hydrolyses (Liu et al., 2003), which opens the middle ring and creates an acidic product. This causes the pH of the medium to drop and in turn immobilizes the daphnids. When the test solution was properly buffered, no effect at all was observed, demonstrating that the effects in the non-buffered experiment were due to the pH rather than to DOPO itself. In the environment it is likely that DOPO concentrations are low enough to be naturally buffered, because it is unlikely that concentrations as high as 100 mg L^{-1} will occur. Therefore we do not expect any acute effect of DOPO on daphnids in a natural environment. Finally, ATO was classified as having a moderate toxicity and TPP as high (Table 1). Both ATO and TPP have very little impurities (<1%) and these are not expected to have caused the observed toxicity. In the pH range 2–10, the solubility of ATO is independent of pH (Filella et al., 2002), thus no significant effects are expected at environmental conditions that deviate from the test conditions.

Reviewing the available ecotoxicity data of HFFRs, we revealed that large data gaps and inconsistent observations are reported on the properties of individual compounds (Waaijers et al., 2013). Thus many of these HFFRs are already being marketed, although their toxicity is virtually unknown. Therefore, the aim of this study was to generate reliable toxicity data for a selection of HFFRs that are suitable replacements for BFRs in polymers. The results showed that DOPO and MHO, for which no experimental data were available until now (Waaijers et al., 2013), exhibit a low acute toxicity to *D. magna*. For ATO a wide toxicity range —from moderate to low toxicity — was reported thus far (e.g. ECHA Database original study, 1998, 1989) and for TPP ranging from moderate to high (Waaijers et al., 2013). The present study narrowed this range down to moderate and high toxicity respectively.

For each of the compounds ALPI, APP, BDP, MPP, ZHS and ZS only a single toxicity report was available from the literature, nominal concentrations were reported only and/or details were lacking on the experimental set-up (Waaijers et al., 2013). For these compounds, the present study generated more reliable toxicity data. For ALPI and BDP the toxicity was higher than reported before (e.g. Australian Government Regulator of Industrial Chemicals, 2000; ECHA Database original study, 2005–Apr–13). Daphnid clone specific differences in sensitivity potentially contribute to the dissimilarities in toxicity studies, such as reported by Baird et al. (1991). Yet the conformity with the literature of the present toxicity data on the reference toxicant $K_2Cr_2O_7$ and BFR TBBPA (EHCA Database original study, 1978; Liu et al., 2007) shows that the presently used clone met the requirement set by the guideline (OECD, 2004). These results, as well as the high mobility in controls and EC_{50} values based on actual concentrations all substantiate the robustness of the toxicity data in the present study. However, it should be kept in mind that although *D. magna* is a standardized and ecologically highly relevant test organism, toxicity is species and compound specific (Posthuma et al., 2001) and other endpoints should be examined as well to complement this study. Such research is currently in progress as part of the European project ENFIRO (2009), in which general in vitro endpoints such as cytotoxicity and more specific ones such as induction of calcium homeostasis (IRAS institute, University of Utrecht, The Netherlands) and of the estrogen receptor (IVM Institute, VU University, Amsterdam, The Netherlands) are being studied.

5. Conclusions

By testing all compounds under the same conditions in an identical set up we provided missing toxicity data of new generation flame retardants that inevitably will be emerging compounds in the aquatic environment. In this way the suitability of HFFRs as alternative FR was evaluated in a reliable manner. The best candidates for BFR replacements are, according to this study: APP, ALPI, DOPO, MHO, MPP, ZS and ZHS. In contrast we indicated that toxic risks exist for ATH, ATO, BDP, RDP and TPP.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.scitotenv.2013.06.110.

References

Adema DMM. *Daphnia magna* as a test animal in acute and chronic toxicity tests. Hydrobiologia 1978;59:125–34.

Alaee M, Arias P, Sjodin A, Bergman A. An overview of commercially used brominated flame retardants, their applications, their use patterns in different countries/regions and possible modes of release. Environ Int 2003;29:683–9.

Australian Government Regulator of Industrial Chemicals. Phosphoric acid, (1-methylethylidene) di-4,1-phenylene tetraphenyl ester (Fyrolflex BDP). File No: NA/773; 2000.

Baird DJ, Barber I, Bradley M, Soares AMVM, Calow P. A comparative-study of genotype sensitivity to acute toxic stress using clones of *Daphnia magna* Straus. Ecotoxicol Environ Saf 1991;21:257–65.

Betts KS. Formulating green flame retardants. Environ Sci Technol 2007;41:7201–2.

Birnbaum LS, Staskal DF. Brominated flame retardants: cause for concern? Environ Health Perspect 2004;112:9–17.

Boon JP, Lewis WE, Tjoen-A-Choy MR, Allchin CR, Law RJ, de Boer J, et al. Levels of polybrominated diphenyl ether (PBDE) flame retardants in animals representing different trophic levels of the North Sea food web. Environ Sci Technol 2002;36:4025–32.

Burns CW. Relation between filtering rate, temperature, and body size in four species of *Daphnia*. Limnol Oceanogr 1969:693–700.

De Schamphelaere KAC, Heijerick DG, Janssen CR. Comparison of the effect of different pH buffering techniques on the toxicity of copper and zinc to *Daphnia magna* and *Pseudokirchneriella subcapitata*. Ecotoxicology 2004;13.

ENFIRO. ENFIRO Webpage. www.enfiro.eu, 2009.

European Chemicals Agency (ECHA) Database. 2,2′,6,6′-tetrabromo-4,4′-isopropylidenediphenol — aquatic toxicity. Short-term to aquatic invertebrates (ecotoxicological information) experimental supporting study 2 original study; 1978.

European Chemicals Agency (ECHA) Database. European Chemicals Agency (ECHA) database (original study 1989) diantimony trioxide — aquatic toxicity. Short-term

to aquatic invertebrates (ecotoxicological information) experimental not specified study 8 original study; 1989.
European Chemicals Agency (ECHA) Database. Diantimony trioxide — aquatic toxicity. Short-term to aquatic invertebrates (ecotoxicological information) experimental not specified study 7 original study; 1998.
European Chemicals Agency (ECHA) Database. 2,2′,6,6′-tetrabromo-4,4′-isopropylidenediphenol — aquatic toxicity. Short-term to aquatic invertebrates (ecotoxicological information) experimental key study 1 original study; 2003.
European Chemicals Agency (ECHA) Database. Aluminium tris(dialkylphosphinate) — aquatic toxicity. Long-term to aquatic invertebrates (ecotoxicological information) experimental key study 1 original study; 2005.. [Apr-13].
European Parliament (E.P.). Report on the joint text approved by the Conciliation Committee for a European Parliament and Council directive amending for the 24th time Council Directive 76/769/EEC relating to restrictions on the marketing and use of certain dangerous substances and preparations (pentabromodiphenyl ether, octabromodiphenyl ether)(PE-CONS 3664/2002 C5-0500/2002 2001/0018(COD)). Report A5-0437/2002; 2002.
European Union. Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH), establishing a European Chemicals Agency. no1907/2006; 2006849.
European Union. Regulation (EC) No 1272/2008 of the European Parliament and of the Council on classification, labelling and packaging of substances and mixtures. no1272/2008; 20081355.
Filella M, Belzile N, Chen Y. Antimony in the environment: a review focused on natural waters: II. Relevant solution chemistry. Earth Sci Rev 2002;59:265–85.
Haanstra L, Doelman P, Voshaar JHO. The use of sigmoidal dose–response curves in soil ecotoxicological research. Plant Soil 1985;84:293–7.
Hem JD, Geological Survey (U.S.). Study and interpretation of the chemical characteristics of natural water. Department of the Interior, U.S. Geological Survey; 1985.
IBM. IBM SPSS Statistics version 20 release 20.0.0 1989; 2011.
Illinois Environmental Protection Agency. Report on alternatives to the flame retardant DecaBDE: evaluation of toxicity, availability, affordability, and fire safety issues — a report to the governor and the general assembly; 2007. p. 86.
Leeuwen CJv, Vermeire TG. Transport, accumulation and transformation processes; bioaccumulation. In: Anonymous, editor. Risk Assessment of Chemicals: An Introduction. Springer; 2007. p. 96.
Liu YL, Wei WL, Chen WY. Reversibility of hydration and dehydration reactions of a phosphaphenanthrene oxide (DOPO) pendent group on polyamide. Polym Int 2003;52:1275–9.
Liu H, Liu X, Wang X, Wang X, Yu H. Toxicity of BPA and TBBPA to *Daphnia magna* and zebrafish *Brachydanio rerio*. Huan Jing Ke Xue 2007;28:1784–7.
Meerts IATM, Letcher RJ, Hoving S, Marsh G, Bergman A, Lemmen JG, et al. In vitro estrogenicity of polybrominated diphenyl ethers, hydroxylated PBDEs, and polybrominated bisphenol A compounds. Environ Health Perspect 2001;109:399–407.
Merck Chemicals - Product Information (MSDS). Aluminium hydroxide fine powder pure, hydrargillite 101093 (CAS 21645-51-5); 2007.
Michaud JP. A citizens' guide to understanding and monitoring lakes and -streams. Washington: The Department of Ecology Publications; 1991.
Organisation for Economic Co-operation and Development (OECD). OECD guideline for testing of chemicals — *Daphnia* sp., acute immobilisation Test 202. Section 2: Effects on Biotic Systems; 2004.
OSPAR. Certain brominated flame retardants — polybrominated diphenylethers, polybrominated biphenyls, hexabromo cyclododecane. (2004 updated)OSPAR Priority Substances Series; 2001.
Posthuma L, Suter GW, Traas TP. Species sensitivity distributions in ecotoxicology. CRC Press; 2001.
Riddell N, Wellington Laboratories. Resorcinol bis(diphenyl phosphate) RDP; 2012a [7 March].
Riddell N, Wellington Laboratories. Bisphenol A bis(diphenyl phosphate) BDP; 2012b [5 March].
Robrock KR, Korytar P, Alvarez-Cohen L. Pathways for the anaerobic microbial debromination of polybrominated diphenyl ethers. Environ Sci Technol 2008;42:2845–52.
SRI Consulting (SRIC). Worldwide FR consumption and geographical distribution; 2004.
Stumm W, Morgan JJ. Aquatic chemistry: chemical equilibria and rates in natural waters; 1996.
U.S. Environmental Protection Agency (EPA). Waste treatment; upgrading metal-finishing facilities to reduce pollutions. EPA 625/3-73-002; 1973273.
U.S. Environmental Protection Agency (EPA). Flame retardants in printed circuit boards (Draft); 2008273.
Waaijers SL, Kong D, Hendriks HS, Wit CA, Cousins IT, Westerink RHS, et al. Persistence, bioaccumulation, and toxicity of halogen-free flame retardants. In: Whitacre DM, editor. Reviews of environmental contamination and toxicology. New York: Springer; 2013. p. 1–71.
Wood JM. Effects of acidification on the mobility of metals and metalloids — an overview. Environ Health Perspect 1985;63:115–9.
Wren CD, Stephenson GL. The effect of acidification on the accumulation and toxicity of metals to freshwater invertebrates. Environ Pollut 1991;71:205–41.